

ROBUST ESTIMATION OF SCALE BASED ON HAMPEL'S *LEMMA FIVE* WITH APPLICATION TO THE RAYLEIGH FAMILY

M. Kidd: Institute for Maritime Technology (Pty.) Ltd.,
P.O. Box 181, Simon's Town, 7995.

and

Nico F. Laubscher: South African Nylon Spinners (Pty.) Ltd., P.O. Box
272, Bellville, 7535.

Key words: Change-of-variance function; Fisher consistency; influence function; MAD; M-estimation; MLE; point estimation; Rayleigh distribution; robust estimation; scale parameter; score function; trimming.

Summary: A robust scale equivariant M-estimator based on Hampel's Lemma 5 is developed for the general one-parameter scale family of distributions. Specific details are worked out for the case where the underlying distribution is a member of the Rayleigh family. The influence and change-of-variance functions are determined as well as an optimal value for its trimming constant, the latter over a range of contaminated distributions, ranging from no and inlier contamination to severe outlier contamination. The estimator is compared with two known and two new robust competitors. Even though there is not much to choose between them, the Hampel estimator seems to have a slight edge over the others on the basis of an average MSE criterion. The estimator is then applied to a data set which conceivably arose from an underlying Rayleigh distribution and is believed to include some outliers. The robustness of the Hampel point estimator is demonstrated by doing the computations with and without these extreme observations.

1. INTRODUCTION

The problem of robust estimation of a one-dimensional scale parameter has been mentioned in the earliest papers on M-estimation (e.g. Huber, 1964,

STMA 1993 SUBJECT CLASSIFICATION: 04:110(04:010)

section 10). However, the location problem received most of the attention on this topic in the literature due to its overriding importance in applications. Details of robust M-estimation of scale (for the Gaussian and the Rayleigh families of distributions), have been presented by Kidd (1991), who developed the necessary theory and described some important practical associated applications. In the present paper we present some general developments in scale estimation based on the Rayleigh distribution as underlying family. If the reader wonders why it is considered necessary to specialize a general theory to such detail as the Rayleigh distribution, it has to be noted that scale estimation cannot be divorced from the parametric family that underlies its structure. In practical application a general theory is of limited use if details belonging to the specific situation are not available.

Our motivation for initiating this theory started with a practical problem in which the biomass of a population of plants had to be estimated. The biomass was related to the density of the plants in the field which in turn was to be estimated through the use of a sampling scheme common in botanical studies, known as the *point centred quarter* sampling method. Under assumptions that were considered reasonable, it appears that the Rayleigh family is central to that estimation problem (cf. Laubscher, 1983). Moreover, our data set contained some suspicious (outlying) observations and thus we had reservations about the application of maximum likelihood estimation in view of its known sensitivity to outliers. However, at that stage no theory of scale M-estimation has been developed for the Rayleigh family and we realized that if we wanted to apply such theory to our data, we would have to derive the required formulas for the Rayleigh! This led to Kidd's (1991) study. The case study that provided this motivation is explored in more detail in section 6.

A further important characteristic of the Rayleigh, and convincing argument for its practical importance, rests on the fact that it is actually a special case of the Weibull distribution (with shape parameter 2). It is characterized by a single scale parameter. This descent strengthens our motivation, due to the recognized importance of the Weibull family and its

derivatives such as the Rayleigh and the exponential in the study of life data.

Apart from the detailed study of scale M-estimation of the Rayleigh family presented here, we observe that the general topic of scale M-estimation has also recently attracted attention through an application in the unexpected field of the design of experiments (cf. Juan and Peña, 1992). However, the Gaussian family underlies that study.

In the scale estimation problem we consider estimation of the one-dimensional scale parameter θ of a family of distributions $F_\theta(x) = F(x/\theta)$, where F is assumed known, but we want to allow for deviations from this known form. In particular we consider the M-estimation methodology where the starting point in determining the estimator is a defining equation $\psi = 0$ (see e.g. Hampel *et al.*, 1986). In his Ph.D. thesis, Hampel (1968) defined a general M-estimator (Lemma 5) based on the ψ -function $\tilde{\psi}(x) := [s_\theta(x) - a]_{-b}^{+b}$. Here $s_\theta(x) := \frac{\partial}{\partial \theta} \ln f_\theta(x)$ is the Fisher score function for the assumed underlying distribution and a is a value, in principle dependent on b and θ , which is selected to ensure Fisher consistency of the estimator. As is the custom, we identify the estimator with its defining ψ -function. The expression $[y]_{-b}^{+b}$, $b > 0$, denotes the *trimming function*. It is defined as y except when y exceeds b , in which case it takes on the value b , or when y does not exceed $-b$ in which case it takes on the value $-b$.

This estimator has the property that it has the smallest asymptotic variance for that class of Fisher consistent M-estimators with gross error sensitivity (supremum of the *influence function*) not exceeding that of $\tilde{\psi}$. This is useful because demanding robustness usually causes an increase in the variance of the estimator. Thus the Lemma 5 estimator is a robust estimator with smallest asymptotic variance relative to other comparable robust estimators.

Applying this theorem to the scale case is more complex and was not previously investigated in depth. The difficulty lies in the fact that the trimming value b is in fact a function of the unknown θ (which can easily

be overlooked when using Hampel's notation). To make things more intuitive, we made $b > 0$ a fixed constant and then adapted the theorem for defining a scale estimator in the following way: adapt the ψ -function by replacing $s_\theta(x)$ by $s(x/\theta)$ where $s(x)$ is the Fisher score function for the standard member of the assumed underlying distribution. Thus ψ is of the form $\psi(x/\theta)$ which ensures the *scale equivariance* of the estimator (i.e. $T_n(x)$ satisfies $T_n(cx) = |c| T_n(x)$ for an arbitrary constant c).

In section 2, we formally rewrite the theorem applied to the general single parameter scale family $F_\theta(x) = F(x/\theta)$. Section 3 highlights the relationship between Hampel's original theorem and the adapted theorem of section 2. In section 4 the theorem is applied to derive a new robust scale estimator for the Rayleigh distribution. This estimator is compared in section 5 with competing scale estimators.

2. ADAPTING HAMPEL'S LEMMA 5 FOR SCALE

We introduce Hampel's *Lemma 5*, specifically for scale, by way of a formal formulation as follows:

Theorem (Hampel's *Lemma 5* for scale). Let $0 < b < \infty$ be some constant, not depending on θ . Define

$$(2.1) \quad a(b) \equiv \left\{ k \mid \int [s(x/\theta) - k]_{-b}^{+b} dF_\theta(x) = 0 \right\}$$

$$(2.2) \quad \chi_\theta(x) \equiv [s(x/\theta) - a(b)]_{-b}^{+b}, \text{ and}$$

$$d_\theta(b) \equiv \int \chi_\theta(y) s_\theta(y) dF_\theta(y).$$

Then the following results hold:

- $a(b)$ exists independently of θ , implying that $\int \chi_\theta(x) dF_\theta(x) = 0$ (Fisher consistency).
- $d_\theta(b) > 0$ (asymptotic variance exists).
- Among all functions ψ_θ that satisfy

$$(2.3) \quad \diamond \quad \int \psi_\theta(x) dF_\theta(x) = 0 \quad (\text{Fisher consistency}),$$

$$(2.4) \quad \diamond \quad \gamma^*(\psi_\theta, b) := \sup_z \left| \frac{\psi_\theta(x)}{\int \psi_\theta(y) s_\theta(y) dF_\theta(x)} \right| \leq c := \frac{b}{d_\theta(b)} < \infty$$

(gross error sensitivity bounded and not exceeding that of χ_θ),

with $\int \psi_\theta(y) s_\theta(y) dF_\theta(y) \neq 0$,

the choice $\psi_\theta(x) = \chi_\theta(x)$ minimizes the asymptotic variance of the

estimator, viz. $\frac{\int \psi_\theta^2(x) dF_\theta(x)}{[\int \psi_\theta(y) s_\theta(y) dF_\theta(y)]^2}$.

Discussion

We now present an explanation assisting in an intuitive understanding of the theorem. Expression (2.2) is the nucleus of the theorem since it defines the ψ -function used to construct the estimator. Looking closer at this expression one finds that it contains the Fisher score function $s(x)$ which ties in with the theory of maximum likelihood estimation. The trimming function trims the score function ensuring that the estimator has a bounded influence

function, making it *B-robust* (cf. Hampel *et al.*, 1986). The function $a(b)$ in (2.1) is selected to make the estimator Fisher consistent, thus ensuring that the correct parameter (and not a scaled version thereof) is estimated. The expression $d_{\theta}(b)$ is the denominator of the gross error sensitivity γ^* , and (2.3) and (2.4) define the class of estimators among which χ_{θ} has minimum asymptotic variance.

Proof

The proof of this theorem is analogous to that of Hampel's *Lemma 5* (cf. Hampel *et al.*, 1986, pp. 117–119). Details for scale have been worked out by Kidd (1991).

3. RELATIONSHIP BETWEEN ADAPTED AND ORIGINAL HAMPEL LEMMA 5

The estimator from Hampel's *Lemma 5*, is defined for a general underlying distribution $F_{\theta}(x)$. The adapted estimator from section 2 is defined specifically for the scale family of distributions $F(x/\theta)$. We now investigate the relationship between the adapted estimator, χ , and the original estimator, $\tilde{\psi}$, as applied to the scale family.

The Hampel *Lemma 5* ψ -function is

$$(3.1) \quad \tilde{\psi}_{\theta}(x) = [s_{\theta}(x) - a_{\theta}(b(\theta))]_{-b(\theta)}^{+b(\theta)}$$

with $a_{\theta}(t) = \{k | \int [s_{\theta}(x) - k]_{-t}^{+t} dF_{\theta}(x) = 0, t > 0\}$, where the dependence of all functions and parameters on θ has been explicitly identified. Choosing $b(\theta) = b/\theta$ for any $b > 0$, it follows that

$a_{\theta}(b/\theta) = \{k | \int [s(x/\theta) - \theta k]_{-b}^{+b} dF(x/\theta) = 0\} = 1/\theta \cdot a(b)$ where $a(b)$ is the Fisher consistency function for the adapted estimator defined in (2.1). Then

$$\tilde{\psi}_{\theta}(x) = [1/\theta \cdot s(x/\theta) - 1/\theta \cdot a(b)]_{-b/\theta}^{+b/\theta} \equiv [s(x/\theta) - a(b)]_{-b}^{+b}$$

which is the adapted Hampel ψ -function, χ , in (2.2).

4. APPLICATION TO THE RAYLEIGH FAMILY

The Rayleigh family, specified below, is an asymmetric member of the general scale family of distributions. It is defined on the positive real line, which makes it special from the point of view of M-estimation, which has by and large been applied to families of underlying symmetric distributions. It underlies an interesting class of practical situations, alluded to in the introduction, where the robust estimation of its parameter is relevant (cf. Kidd, 1991). Its density function is $r_{\theta}(x) := (1/\theta^2) \cdot 2x \cdot \exp(-(x/\theta)^2)$, with $x, \theta \in \mathbb{R}^+$. The distribution and density functions of its standardized member are $R(x) = 1 - \exp(-x^2)$ and $r(x) = 2x \exp(-x^2)$ respectively. The Fisher score functions for the Rayleigh and its standard member are

$$s_{\theta}(x) = \frac{\partial}{\partial \theta} \ln(r_{\theta}(x)) = (2/\theta)[(x/\theta)^2 - 1], \text{ and } s(x) = 2(x^2 - 1) \text{ respectively.}$$

The ψ -function for this M-estimator is $[2(x/\theta)^2 - 2 - a(b)]_{-b}^{+b}$, $a(b)$ being defined implicitly through $\{a | \int_0^{\infty} [2x^2 - 2 - a]_{-b}^{+b} r(x) dx = 0\}$. The integral to be evaluated $I = \int_0^{\infty} [2x^2 - 2 - a]_{-b}^{+b} r(x) dx$, may be simplified as follows:

$$I = \begin{cases} e^{-k_1^2(b+2k_1^2-a)+} e^{-k_2^2(b-2k_2^2+a)-b} & \text{for } a+b+2 > 0, a-b+2 > 0 \\ e^{-k_2^2(b-2k_2^2+a)-a} & \text{for } a+b+2 > 0, a-b+2 < 0 \\ b & \text{for } a-b+2 < 0 \end{cases}$$

where $k_1 := \sqrt{\frac{1}{2}(a-b+2)}$ and $k_2 := \sqrt{\frac{1}{2}(a+b+2)}$. For a random sample $X_1; \dots; X_n$, the robust estimator T_n for θ is

$$T_n(X_1; \dots; X_n) := \left\{ \theta \left| \sum_{i=1}^n \left[2(X_i/\theta)^2 - 2 - a(b) \right]_{-b}^{+b} = 0 \right. \right\}.$$

For further referencing we call this estimator the *HampelR* estimator, the last character identifying the Rayleigh as the underlying family.

Influence function and change-of-variance function

To determine the influence function and change-of-variance function, we first introduce notation $A(\chi) := \int \chi^2(x) dR(x)$ and $B(\chi) := \int x\chi'(x) dR(x)$. These help to simplify some of the following expressions. Note that χ' does not necessarily exist in every x , but this problem has been circumvented by Hampel *et al.* (1986, p. 127) through the use of δ -functions. The *influence function* for the M-estimator can be written as (see Hampel *et al.*, 1986)

$$IF(x; \chi; R) = \frac{\chi(x)}{B(\chi)}$$

and the *change-of-variance function* as

$$CVF(x; \chi, R) := \frac{A(\chi)}{B^2(\chi)} \left[1 + \frac{\chi^2(x)}{A(\chi)} - 2 \frac{x\chi'(x)}{B(\chi)} \right].$$

Expressions for $A(\chi)$ and $B(\chi)$ can be derived (using tedious algebraic manipulation) as

$$A(\chi) = \begin{cases} b^2 + \left[(2k_1^2 - a)^2 - b^2 + 4 \right] e^{-k_1^2} & \\ - \left[(2k_2^2 - a)^2 - b^2 + 4 \right] e^{-k_2^2} & \text{for } a + b + 2 > 0, a - b + 2 > 0 \\ a^2 + 4 - \left[(2k_2^2 - a)^2 + 4 - b^2 \right] e^{-k_2^2} & \text{for } a + b + 2 > 0, a - b + 2 < 0 \\ b^2 & \text{for } a + b + 2 < 0 \end{cases}$$

where $k_1 := \sqrt{\frac{1}{2}(a-b+2)}$ and $k_2 := \sqrt{\frac{1}{2}(a+b+2)}$ and

$$B(\chi) = \begin{cases} 4(k_1^2 + 1)e^{-k_1^2} - 4(k_2^2 + 1)e^{-k_2^2} & \text{for } a + b + 2 > 0, a - b + 2 > 0 \\ 4 - 4(k_2^2 + 1)e^{-k_2^2} & \text{for } a + b + 2 > 0, a - b + 2 < 0 \\ 0 & \text{for } a + b + 2 < 0 \end{cases}$$

Graphs of both functions are depicted in figure 1.

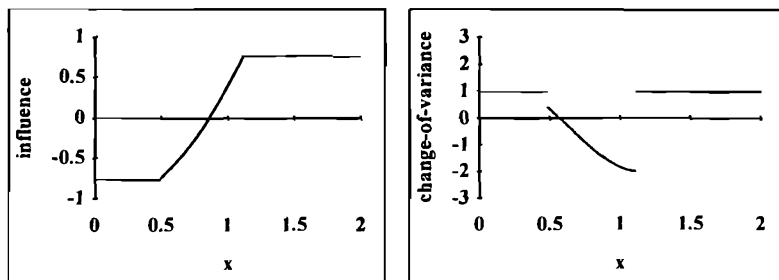


Figure 1: Influence function and change-of-variance function of the Rayleigh family

From figure 1 we see that the influence and change-of-variance functions are bounded, which implies that the estimator is both *B-robust* and *V-robust*.

The following table gives the Fisher consistency constant $a(b)$, asymptotic efficiency, gross-error sensitivity, and change-of-variance sensitivity for various choices of the tuning constant b .

Table 1: Various performance measures for different choices of b

b	$a(b)$	Asymp- totic variance	Asymp- totic efficiency	Gross-error sensitivity	Change-of- variance sensitivity
0,1	-0,613	0,504	0,496	0,722	2,034
0,5	-0,593	0,447	0,559	0,732	2,199
1	-0,531	0,395	0,633	0,766	2,488
2	-0,317	0,324	0,772	0,910	3,559
3	-0,180	0,290	0,862	1,081	5,024
4	-0,105	0,274	0,912	1,261	6,814
5	-0,062	0,264	0,947	1,452	8,983
6	-0,037	0,259	0,965	1,654	11,575
7	-0,022	0,255	0,980	1,865	14,626
8	-0,014	0,253	0,988	2,085	18,164
∞	0	0,25	1	∞	∞

Finding an optimal tuner

To determine the choice of the tuning constant, a Monte Carlo simulation study was done. Three factors were considered, namely

- Sampling distribution.
- Sample size, n .
- Tuning constant.

All samples were randomly drawn from the standard member of the Rayleigh distribution (denoted by $R(1)$), contaminated in the following ways (with abbreviated notation):

I	No contamination	$0\% R(1)$
II	10% with $R(2)$	$10\% R(2)$
III	20% with $R(2)$	$20\% R(2)$
IV	10% with $R(5)$	$10\% R(5)$
V	20% with $R(5)$	$20\% R(5)$
VI	20% with $R(0,1)$	$20\% R(0,1)$

Cases II and III represent slight outlier contamination whereas IV and V represent severe outlier contamination. The last contamination, VI, represents *inlier* contamination. Only one inlier-contaminated distribution was included since in the case of the Rayleigh (where the scale parameter is the only one of interest and does not play a secondary role, e.g. in standardizing) it is not considered to be as serious as outlier contamination. These choices thus attempt to represent a wide spectrum of contaminations that may be encountered in practice.

The sample sizes chosen were $n = 20, 30$ and 40 and all simulations were done for $\theta = 1$. This single choice is sufficient in view of the scale equivariance of the estimator. The mean squared error (MSE) was selected as criterion for comparing the estimators, and was calculated as follows:

- A Monte Carlo study with $N = 1\,000$ repetitions produced a thousand different outcomes of the estimator under study, namely $\hat{\theta}_1 \dots \hat{\theta}_N$.

- The mean and variance of these outcomes were calculated as

$$\bar{\theta} := \frac{1}{N} \sum_{i=1}^N \hat{\theta}_i \quad \text{and} \quad s^2 := \frac{1}{N-1} \sum_{i=1}^N (\hat{\theta}_i - \bar{\theta})^2.$$

- The mean square error of T_n can be written (in terms of its variance and bias) as $\text{MSE}[T_n; \theta] = \text{VAR}[T_n] + (\theta - E[T_n])^2$, where θ is the true value of the scale parameter and it is estimated by $\widehat{\text{MSE}}[T_n; \theta] = s^2 + (\theta - \bar{\theta})^2$.

Figure 2 depicts a typical result of the simulation study for the *HampelR* estimator with sample size of $n = 40$.

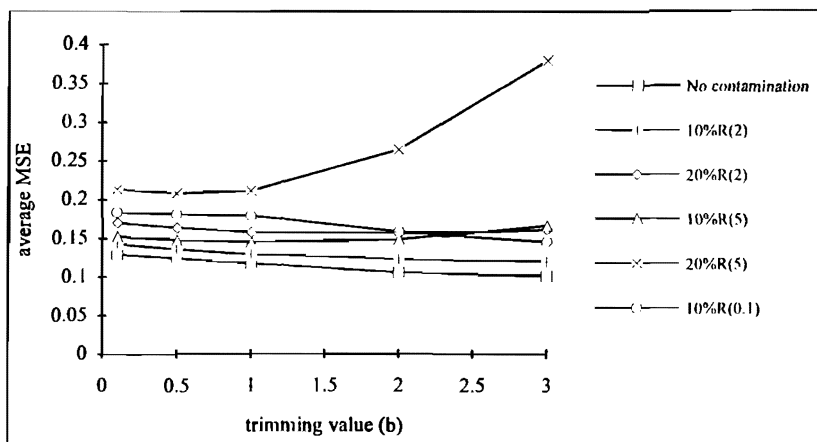


Figure 2: Mean square errors of the *HampelR* estimator for $\theta = 1$ and sample size 40.

To be able to choose an optimal trimming value over the various forms of contamination, the MSEs had to be averaged over the different distributions. Our viewpoint is that the heavy contaminations should not

carry too much weight in our decision because in practice when one has a heavily contaminated distribution, it might be better to review the underlying assumptions. We therefore decided to use the *Huber 1.5* robust estimator (Huber, 1964) which will downweigh the effect of the heavy contaminations, but still take them into account. Figure 3 depicts the results for the *HampelR* estimator. From the graph we selected the tuning constant that minimizes the robust average MSE and the choice $b = 1$ was made.

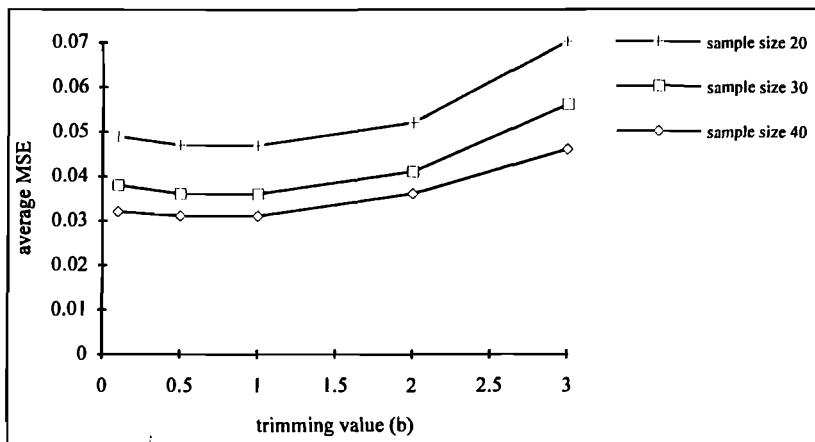


Figure 3: Average MSEs of the *HampelR* estimator for various choices of the tuning constant.

5. COMPARING THE *HAMPELR* TO OTHER SCALE ESTIMATORS

To compare the performance of the *HampelR* estimator with its most important competitors, further Monte Carlo work was done. The maximum likelihood estimator (MLE) for θ , based on a sample of size n , is

$\hat{\theta} = \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2}$. This estimator is quite non-robust, since outliers will

contribute according to their squared values. It nevertheless has a role to play as a reference estimator and would have been the estimator of choice before the advent of robust procedures. The other robust estimators that are included in the study are the following:

MAD(MadR)

This estimator was studied by various authors (see, e.g. Hampel *et al.*, 1986). It is an M-estimator, with estimating equation $\psi_{\text{MAD}}(x) = \text{sign}(x-a)$ where a is a constant ensuring the Fisher consistency of the estimator. The estimator can be written as:

$$T_n = 1/a \cdot \text{Med}(X_i) \text{ with } a = \sqrt{-\ln(1/2)} = 0,8326.$$

This estimator differs from the better-known class of MAD estimators (where the median of the absolute deviations from the median is calculated) in that it is actually nothing more than a scaled median. We have decided to still call it a MAD estimator because it is derived from the same ψ -function as the classical MAD and therefore belongs to the same class.

Cauchy estimator (CauchyR)

Using the Fisher score function of the Cauchy distribution (it is a bounded function), we construct an M-estimator with estimating equation:

$$\psi_c(x) = 2x^2/(1+x^2) - a$$

where $a = 0,8073$ is the Fisher consistency constant. This estimator was introduced by Kidd (1991).

Shortest half (ShortR)

This estimator (introduced by Rousseeuw and Leroy, 1988) is defined as

$$T_n(X_1; \dots; X_n) := c \cdot \min\{X_{(h)} - X_{(1)}; X_{(h+1)} - X_{(2)}; \dots; X_{(n)} - X_{(n-h+1)}\}$$

where $h = [n/2] + 1$, $c = 1,6100$ is the Fisher consistency constant, and $[n/2]$ denotes the integer part of $n/2$.

Trimmed maximum likelihood (TrMleR)

This estimator (see Kidd 1991) is a trimmed version of the MLE and is defined as:

$$T_n := \frac{1}{a} \sqrt{\frac{1}{n - 2[\alpha n]} \sum_{i=[\alpha n]+1}^{n-[\alpha n]} X_{(i)}^2}$$

with $\alpha < 0,5$ and Fisher consistency constant a :

$$a = \sqrt{[1 - \ln(1 - \alpha)]/(1 - 2\alpha) + \alpha[\ln(1 - \alpha) + \ln(\alpha) - 2]/(1 - 2\alpha)}.$$

Based on studies done by Kidd (1991), α was chosen as 0,4.

To compare the *HampelR* estimator with those identified above, a study similar to that described in section 4 was done. The results are depicted in figure 4.

From this we deduce that there is very little to choose between the *HampelR*, *TrMleR* and the *MadR* estimators.

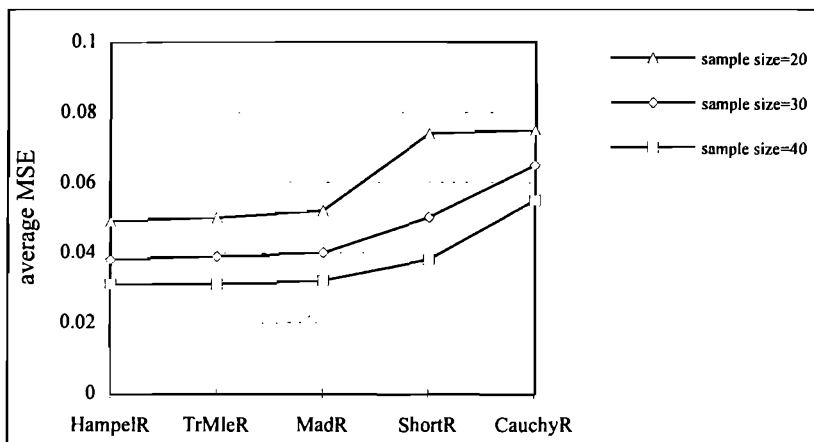


Figure 4. A comparison of estimators for the scale parameter of the Rayleigh distribution.

6. APPLICATION

As an example of a practical application we briefly introduce the *Copiapoa solaris* data sample (see Kidd 1991). *Copiapoa solaris* is a species of Chilean cactus and the problem was to determine the density (number of plants per hectare) of a large growth of these cacti in the Atacama desert. The data set was gathered by the botanist P.J. Weisser (1973). A random starting point was chosen on a random direction through the field of interest. Thereafter samples were obtained at fixed distances (sufficiently far apart to ensure non-overlapping measurements) on the line transect by measuring the distance to the nearest plant in each of the four direction quadrants. Forty independent stopping points led to 160 distance measurements.

Under the assumption of spatial randomness of the plants, it can be derived that such a set of nearest distances arises from the Rayleigh distribution and the density of the plants can be expressed as a function of

the Rayleigh scale parameter (cf. Laubscher, 1983). Probability plots indicated a reasonably good fit with the possible presence of two outliers. Table 2 gives the estimates yielded by the different point estimators for the full data set and for the two outliers discarded. The ShortR estimator was not included here due to the small-sample behaviour of its bias (see Rousseeuw and Leroy 1988).

Table 2: Estimates of the plant density (plants per hectare) from the *C. solaris* data.

Estimator	Estimate (plants/ha)	
	Full dataset	Outliers discarded
MLE	339	391
CauchyR	380	394
HampelR	385	393
MadR	383	400
TrMleR	390	398

From table 2 we see that for the full dataset, the MLE deviates substantially from the rest. All estimates are relatively close together once the outliers are discarded. This example shows the serious effect outliers can have on the MLE and how important it is to recognize and use robust estimators for protection against the unwarranted effect of outliers.

REFERENCES

- HAMPEL, F.R. (1968). *Contributions to the theory of robust estimation*. Ph.D. dissertation. University of California, Berkeley.
- HAMPEL, F.R., ROUSSEEUW, P.J., RONCHETTI, E.M. & STAHEL, W.A. (1986). *Robust Statistics: The approach based on influence functions*. New York: John Wiley.

- HUBER, P.J. (1964). Robust estimation of a location parameter. *Ann. Math. Statist.*, **35**, 73–101.
- JUAN, J., & PEÑA, D. (1992). A simple method to identify significant effects in unreplicated two-level factorial designs. *Communications in Statistics - Theory and Methods*, **21**(5), 1383–1403.
- KIDD, M. (1991). *Robust estimation for scale*. Ph.D. dissertation. University of Stellenbosch, Stellenbosch.
- LAUBSCHER, N.F. (1983). Likelihood analysis of line transect samples. In *Proceedings of the 44th Session of ISI*, Vol. I, pp. 27–30. Madrid: International Statistical Institute.
- ROUSSEEUW, P.J. & LEROY, A.M. (1988). A robust scale estimator based on the shortest half. *Statistica Neerlandica*, **42**, 103–116.
- WEISSER, P.J. (1973). Population measurements of *Pilocopiapoa* at El Cobre, Atacama desert. Manuscript, National Botanical Research Institute, Pretoria.